Project Proposal Form
Institute of Information Technology (IIT)
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Project Description (with in 200 words) :

BiasFree News: Making Bengali News Trustworthy and Fair

In the digital age, misleading headlines and media bias distort truth, making it difficult for readers to trust news. BiasFree News tackles this by debiasing Bengali news articles to enhance accuracy and objectivity

Debiasing Content
Articles often contain biased language through politically charged adjectives, misleading verbs, and sensational phrases. BiasFree News automatically identifies these biased terms and replaces them with neutral equivalents, maintaining original meaning while removing political slant or emotional manipulation. Similar work on bias detection and neutralization in news has been explored in prior research. 1

Accurate Title Generation
Headlines crafted for attention often misrepresent article content. BiasFree News generates accurate, content-aligned titles to prevent reader deception.Related research on Bengali news headline generation using auxiliary information demonstrates how such approaches can improve headline quality and alignment with article content. 2

Although the platform doesn't generate new content, it significantly improves credibility by making news fairer and more accurate. Through automated debiasing, readers access less manipulative content, ensuring trustworthy news engagement.

References:

1. **Raza, S., Reji, D. J., & Ding, C. (2022). Dbias: detecting biases and ensuring fairness in news articles. *International Journal of Data Science and Analytics*.**

2. **Akash, A. U., Nayeem, M. T., Shohan, F. T., & Islam, T. (2023). Shironaam: Bengali News Headline Generation using Auxiliary Information. *EACL 2023*.**

Component	Tool/Technology
Frontend	**React (JavaScript)**
Backend	**FastAPI (Python)**
ML Framework	**Hugging Face Transformers**
NLP Framework	**spaCy, TextBlob, Hugging Face Tokenizers**
Bias Mitigation Method	**Word Replacement, Fairness Constraints, Attention Mechanism**
ML Model	**BERT, RoBERTa (for text classification)**
DL Model	**T5, XLNet, DeBERTa**
Dataset	**Custom Dataset from Bengali News Sources**
Preprocessing Tool	**NLTK, spaCy**
Version Control	**Git, GitHub**
Database	**PostgreSQL**
Environment	**Visual studio/ Jupyter notebook**

Supervisor's Name:_____

Signature of the supervisor:_____

Date:_____

Proposal Presentation Feedback:

Midterm Presentation Feedback: